

## Equilibrium params / Radiation Integrals / RF params

|                                              |          |
|----------------------------------------------|----------|
| Energy [GeV]                                 | 2.86     |
| eq_nemitt_x [ $\mu\text{m}\cdot\text{rad}$ ] | 54.06123 |
| eq_nemitt_y [ $\mu\text{m}\cdot\text{rad}$ ] | 0        |
| eq_nemitt_zeta [m]                           | 0.02705  |
| sigma_delta [%]                              | 0.10403  |
| sigma_zeta [mm]                              | 4.64542  |

|                        |         |
|------------------------|---------|
| I1 [m]                 | 1.44552 |
| I2 [1/m]               | 2.9178  |
| I3 [1/m <sup>2</sup> ] | 0.52168 |
| I4 [1/m]               | 0.02287 |
| I5 [1/m]               | 0.00233 |

|                                 |             |
|---------------------------------|-------------|
| Vrf [MV]                        | 10          |
| frf                             | 0.99995     |
| h                               | 1298        |
| phi_s [deg]                     | 164.043282  |
| eta                             | 3.70E-03    |
| Es [eV]                         | 2.86E+09    |
| $\Delta E_{\text{sep}}$ [keV]   | 47911.84063 |
| $\Delta\delta_{\text{sep}}$ [%] | 1.675239    |

| Parameter                         | Symbol                 | Unit              | Madx                  | Xsuite                       |
|-----------------------------------|------------------------|-------------------|-----------------------|------------------------------|
| Energy                            | E                      | GeV               | 2.86                  | 2.86                         |
| Circumference                     | C                      | m                 | 389.15                | 389.15                       |
| Bunches per train                 | Nb                     |                   | 156                   | -                            |
| Bunch population                  |                        | 10 <sup>9</sup>   | 4.5                   | -                            |
| Bunch spacing                     | tb                     | ns                | 1                     | -                            |
| Basic cell type                   |                        |                   | TME                   | TME                          |
| Number of dipoles                 | Nd                     |                   | 38                    | 38                           |
| Dipole field                      | Ba                     | T                 | 1.2                   | 1.2                          |
| Tunes (hor./ver./sync.)           | Qx/Qy/Qs               |                   | 16.39 / 12.26 / 0.051 | 16.39 / <b>12.27</b> / 0.051 |
| Natural chromaticity (hor./vert.) | $\xi_x/\xi_y$          |                   | -18.99 / -22.85       | <b>-17.45 / -19.37</b>       |
| Normalized horiz. emittance       | $\gamma\epsilon_0$     | $\mu\text{m-rad}$ | 53.89                 | <b>54.06</b>                 |
| Damping times                     | $\tau_c/\tau_y/\tau_z$ | ms                | 2.68 / 2.68 / 1.34    | <b>2.72 / 2.70 / 1.35</b>    |
| Momentum compaction factor        | $\alpha_c$             | 10 <sup>-3</sup>  | 3.72                  | <b>3.70</b>                  |
| Energy loss/turn                  | U0                     | MeV               | 2.75                  | 2.75                         |
| RF voltage                        | Vrf                    | MV                | 10                    | 10                           |
| RF acceptance                     | $\epsilon_{rf}$        | %                 | 1.7                   | <b>1.68</b>                  |
| RF frequency                      | frf                    | GHz               | 1                     | 1                            |
| Harmonic number                   | h                      |                   | 1298                  | 1298                         |
| Energy spread (rms)               | $\sigma_\delta$        | %                 | 0.1                   | 0.10                         |
| Bunch length (rms)                | $\sigma_s$             | mm                | 4.6                   | <b>4.65</b>                  |
| Number of wigglers                | Nwig                   |                   | 36                    | 36                           |
| Wiggler peak field                | Bw                     | T                 | 1.9                   | 1.9                          |
| Wiggler length                    | Lwig                   | m                 | 3                     | 3                            |
| Wiggler period                    | $\lambda_w$            | cm                | 30                    | 30                           |

- [DampingRings < CLIC < TWiki](#)